

1. Understood the fundamental concepts of single-view metrology and how real-world distances can be estimated from a single image using principles of projective geometry.
2. Understood how parallel lines in the real world appear to converge in an image due to perspective projection how vanishing point can be obtained.
3. Learned how to identify and annotate collinear points in an image and why collinearity is essential for applying cross-ratio based distance estimation.
4. Understood the concept of cross-ratio as a projective invariant and how it remains unchanged under perspective transformations.
5. Learned how a known reference distance can be used together with cross-ratio invariance to estimate unknown real-world distances.
6. Understood the significance of selecting an appropriate reference point within a detected bounding box, such as the ground-contact point, for accurate geometric measurements.
7. Learned how image coordinates can be converted into meaningful geometric relationships through projective geometry rather than conventional Euclidean distance calculations.
8. Understood the role of homogeneous coordinates and projective representations in modeling points at infinity and vanishing points.
9. Learned how inaccuracies in vanishing point estimation, object localization, and manual annotation can propagate into the final distance estimates.
10. Experienced combining computer vision techniques with geometric reasoning to solve real-world measurement problems.